

This skills check covers pre-requisite material needed for the first unit of the course, areas and volumes.

1. Identifying Bounded Regions

- a) Shade the region bounded by the following curves: $h(x) = \ln(x)$, $x = 1$, $y = 2$
- b) Shade the region bounded by the following curves: $f(x) = \sin(x)$, $g(x) = e^x$, $x = 0$, $x = 1$
- c) Shade the region bounded by the following curves: $x = -y^2$, $x = -4$

2. Finding Points of Intersection

- a) Find the points of intersection of the curves: $y = x^2 + x$ & $y = 2x + 6$
- b) Find the points of intersection of the curves: $f(x) = \ln(2x)$ & $g(x) = 0$
- c) Find the points of intersection of the curves: $y = \sqrt{2x + 7}$ & $y = x + 2$

3. Definite Integrals for Rotations

- a) Evaluate the definite integral: $\int_0^3 (x + 2)^2 dx$
- b) Evaluate the definite integral: $\int_1^0 (\pi e^t - 1) dt$
- c) Evaluate the definite integral: $\int_1^2 \pi \left(x^2 - \frac{1}{x} \right) dx$

4. U-Substitutions with Definite Integrals

- a) Use a u-substitution to evaluate the definite integral: $\int_0^{\pi/4} \frac{\sin(x)}{\cos(x)} dx$
- b) Use a u-substitution to evaluate the definite integral: $\int_0^1 \frac{e^x}{e^x + 1} dx$
- c) Use a u-substitution to evaluate the definite integral: $\int_1^2 x\sqrt{x^2 - 1} dx$

5. Solving Literal Equations

- a) Solve the equation for x : $y = \frac{2}{x + 3}$
- b) Solve the equation for x : $y = 2\sqrt[3]{x + 1}$
- c) Solve the equation for x : $y = \ln(x + 2)$